

Jayesh Patel

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PROFESSIONAL SUMMARY

Full-stack developer and Computer Science graduate with hands-on experience in web development, mobile development and data science. Proven track record building high-precision computation engines and open-source libraries published on Packagist and PyPI. Adept at delivering clean, maintainable code and integrating AI APIs into production systems.

TECHNICAL SKILLS

Languages: Python, PHP, JavaScript, Java, Go, C, C++, Dart
Frontend: React, Next.js, Vue.js, AngularJS, Tailwind CSS, Bootstrap, SCSS, WebGL, HTML, CSS
Backend: Laravel, Node.js, Express.js, FastAPI, Flask, Django, JWT
Databases: MySQL, PostgreSQL, SQLite, MongoDB
Mobile: Flutter, Android (WebView/Java)
DevOps & Tools: Git, GitHub, GitLab, Nginx, Postman, Vite, VS Code, Linux, PowerShell
AI/ML & Data Science: PyTorch, TensorFlow, Keras, OpenCV, scikit-learn, Pandas, HuggingFace, NLP
AI Media & Generative: Stable Diffusion, GFPGAN, RealESRGAN, CodeFormer, UVR
Design & Creative: Figma, Adobe Creative Cloud, Canva
Other: FFI, ctypes, REST API design, SQLAlchemy, Alembic, AI API integration

WORK EXPERIENCE

Shreesofttech Dec 2024 – Present
Laravel Developer Hybrid

WRTeam May 2024 – June 2024
PHP Developer Intern Remote

- Engineered and maintained a PHP web application, improving codebase efficiency by 50%+ through dynamic fetching optimizations that significantly reduced file count and code volume.
- Collaborated on integrating front-end and back-end systems, accelerating delivery timelines.

CodSoft Jun 2024 – Jul 2024
Flutter Developer Intern Remote

- Built 4 cross-platform Flutter applications covering diverse domains, enhancing user engagement through polished UI and smooth navigation.
- Conducted code reviews and optimized app performance across Android and iOS targets.

CodSoft May 2024 – Jun 2024
Python Programming Intern Remote

- Developed 5 Python applications spanning user management, game logic, security, arithmetic, and task management, implementing 3 new features that improved robustness by 30%.

CodSoft May 2024 – Jun 2024
Data Science Intern Remote

- Analyzed 5 datasets and produced 55 visualizations using Pandas, Matplotlib, and Seaborn to surface actionable insights.
- Trained and evaluated predictive ML models achieving 85%+ accuracy.

PROJECTS

Panchang Core [GitHub](#)
PHP, FFI, Swiss Ephemeris

- Architected a high-precision Hindu calendar engine using a Swiss Ephemeris C-FFI bridge and IEEE 754 doubles to compute Tithi, Nakṣatra, Yoga, Karaṇa, and 30 Muhūrtas with 0.001 arcsecond accuracy.

- Implemented KalaNirnaya logic orchestrating 237 regional festivals (Smarta/Vaishnava) across Amanta/Purnimanta calendars with English, Hindi, and Gujarati output.
- Designed a framework-agnostic architecture deployable as a standalone CLI or as a Laravel package via native Facades and Service Providers.

Swiss Ephemeris FFI

[GitHub \(PHP\)](#) | [GitHub \(Python\)](#)

PHP, Python, FFI, ctypes, C

- Published zero-abstraction 1:1 FFI libraries for the Swiss Ephemeris C API (106 functions, 348 constants) as two sibling packages: PHP via ext-ffi and Python via ctypes, both on Packagist and PyPI.
- Bundled prebuilt .so/.dylib/.dll binaries for Linux, macOS, and Windows (x64/arm64) directly in each package wheel, eliminating manual C compilation for end users.

Hindu Scriptures

[GitHub](#) | [Live demo](#)

Node.js, Express.js, JavaScript, Google Gemini AI, Java

- Built a full-text scripture search platform with AI-powered search via Google Gemini 2.5 Flash API and URL-rewriting-based multilingual translation through Google Translate.
- Delivered both a web app and an Android WebView app, serving original Devanagari content with on-demand translation to multiple languages.

Enhanced Precision Grayscale Converter

[GitHub \(Web\)](#) | [Live demo](#)

Python, FastAPI, Tailwind CSS, JavaScript, Axios

- Engineered a FastAPI backend supporting 7 color-science modes (Rec.709, BT.601, BT.2100, HSL, HSV, L*a*b*, Gamma) with 8/16-bit output in PNG, JPEG, HEIC, TIFF, WEBP, and BMP.
- Developed a responsive frontend with Axios-powered live previews, drag-and-drop, and EXIF/ICC preservation toggles; fully self-hosted with no third-party uploads.

Laravel Gemini AI Translation Extractor

[GitHub](#)

PHP, Laravel, Google Gemini AI, spatie/fork

- Developed an Artisan command that scans Blade/PHP/JS/TS sources for translation keys, generates Laravel PHP & JSON language files via Gemini 2.0 Flash-Lite, and runs concurrent fork-based requests with auto-retry on rate limits.

PostalKit

[GitHub](#)

Python, ctypes, FFI, libpostal

- Delivered a strict 1:1 ctypes wrapper for the OpenVenues libpostal C library (46 functions, 10 structs, 42 flags) with first-run auto-download of ~2 GB ML models and OS-specific binaries, bypassing all manual toolchain setup.

Literary & Linguistic Poetry Analyzer

[GitHub](#)

Python, FastAPI, PyTorch, HuggingFace

- Designed an async FastAPI backend orchestrating 1.5 GB+ HuggingFace Transformer models, spaCy, NLTK, and Stanza for multilingual sentiment, emotion, and phonological classification.
- Implemented SQLAlchemy ORM connection pooling for SQLite, PostgreSQL, and MySQL/MariaDB, with Alembic schema migrations for production-ready scalability.

Crop & Nutrient Recommendations Apps

[GitHub \(Crop\)](#) | [GitHub \(Nutrient\)](#)

Python, Flask, Google Gemini AI, HTML, CSS, JavaScript

- Built two Flask applications powered by Google Gemini 2.0 Flash Lite: one recommending crops by factoring in weather, soil, and water needs; the other generating personalized daily caloric and nutrient targets from user biometric inputs.

Recipe App

[GitHub](#)

Flutter

- Built a cross-platform Flutter recipe browser with search, filtering, and a user-friendly detail view.

Portfolio

[GitHub](#) | [Live demo](#)

HTML, CSS, JavaScript, Tailwind CSS

- Designed and deployed a personal portfolio showcasing projects, skills, and professional background.

EDUCATION

Parul University

B.Tech in Computer Science and Engineering

CGPA: 7.29

Vadodara, Gujarat, India

2021 – 2025

CERTIFICATIONS

- [PHP Developer](#)
- [Python Programming](#)
- [Data Science](#)
- [Flutter Developer](#)
- [Data Analytics with Python](#)
- [Laravel](#)

SOFT SKILLS

Analytical Thinking, Problem Solving, Attention to Detail, Research, Time Management, Adaptability, Communication, Teamwork, Leadership